## 2.2 Project goal and requirements

When designing software with specific functionality, we typically start with the problem statement and document its requirements.

### 2.2.1 Goal: Build a simple interactive AI chat interface

Our objective is to create a web application where users can

- Ask questions in natural language
- Receive intelligent responses from an AI agent
- View and scroll through their entire conversation history

Figure 2.2 shows how these interactions relate and the flow started by a user interaction.

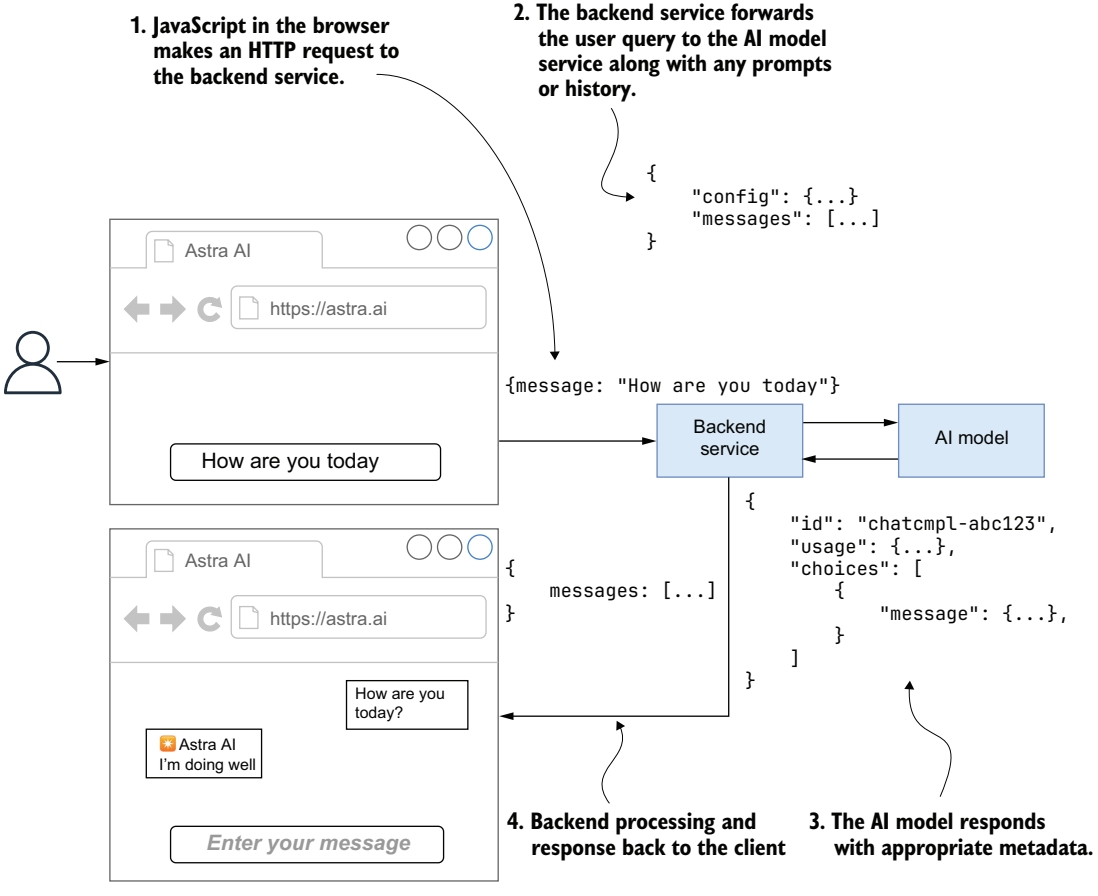



**Figure 2.2 The user interaction flow in the chat application. The user types a question and presses Enter. The web UI submits the question to the backend service, which processes it and sends it along with context to the AI model. The AI model responds with a query response, which the backend service verifies and sends back to the web UI to update the conversation messages. This cycle allows for follow-up questions.**